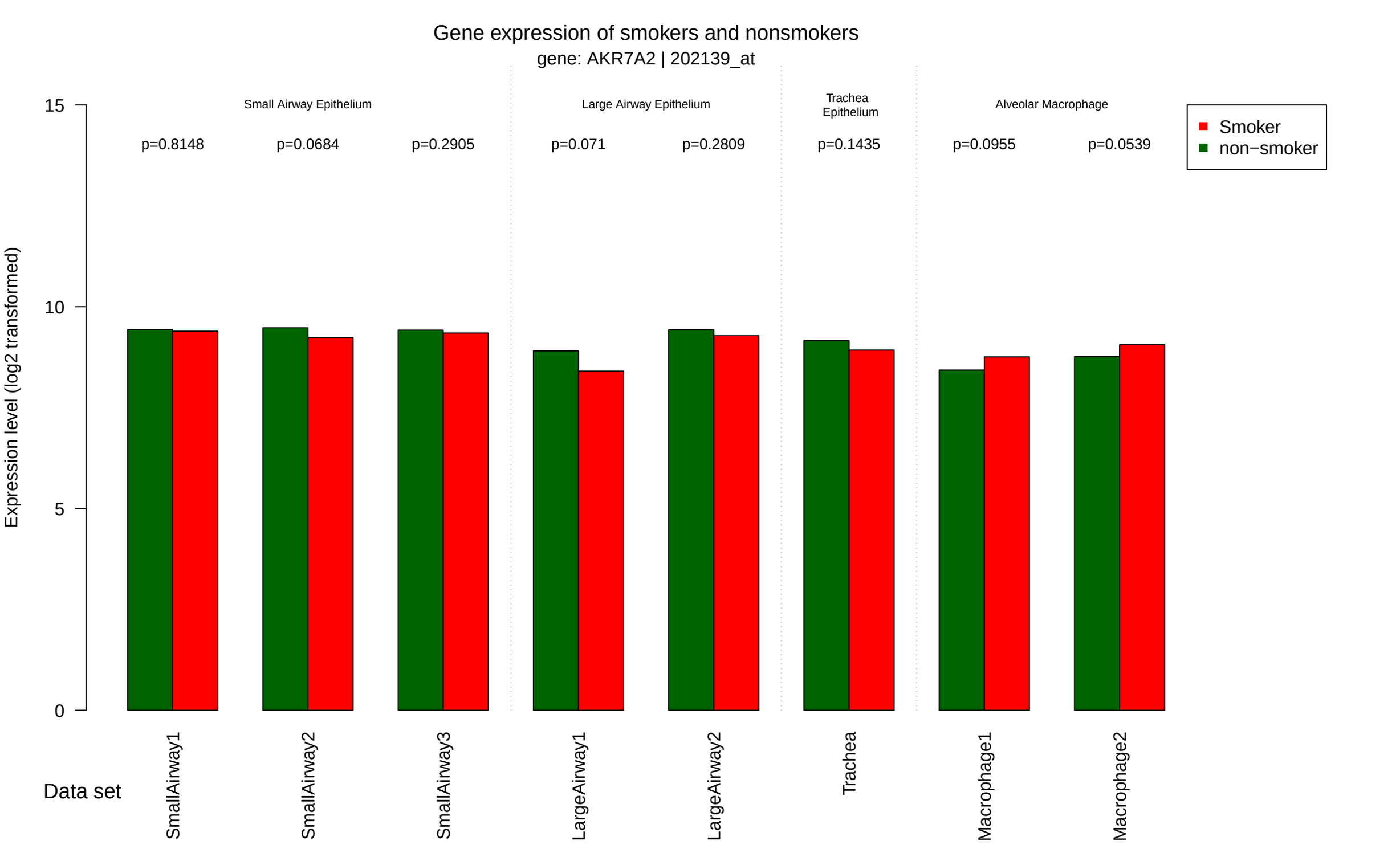
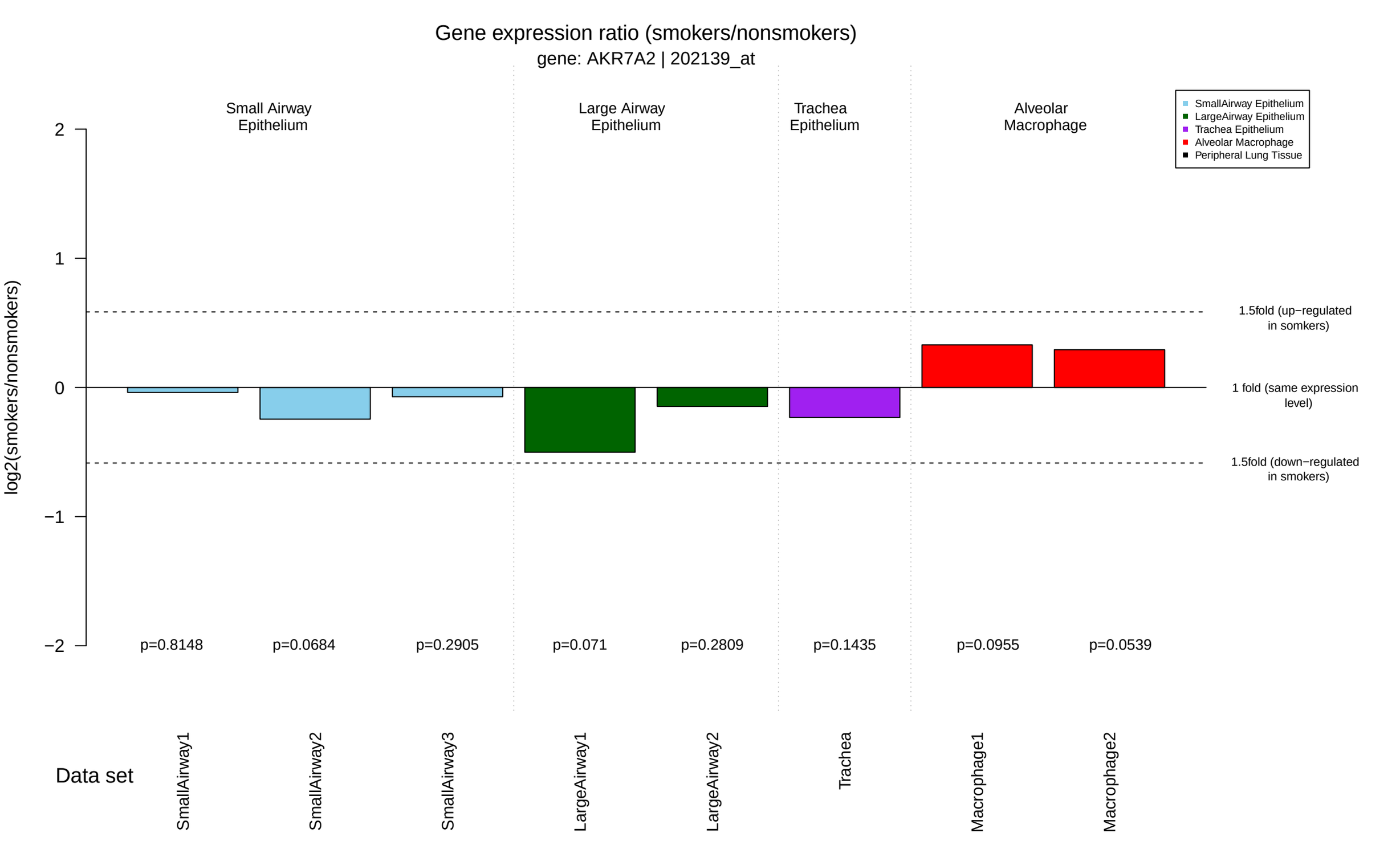
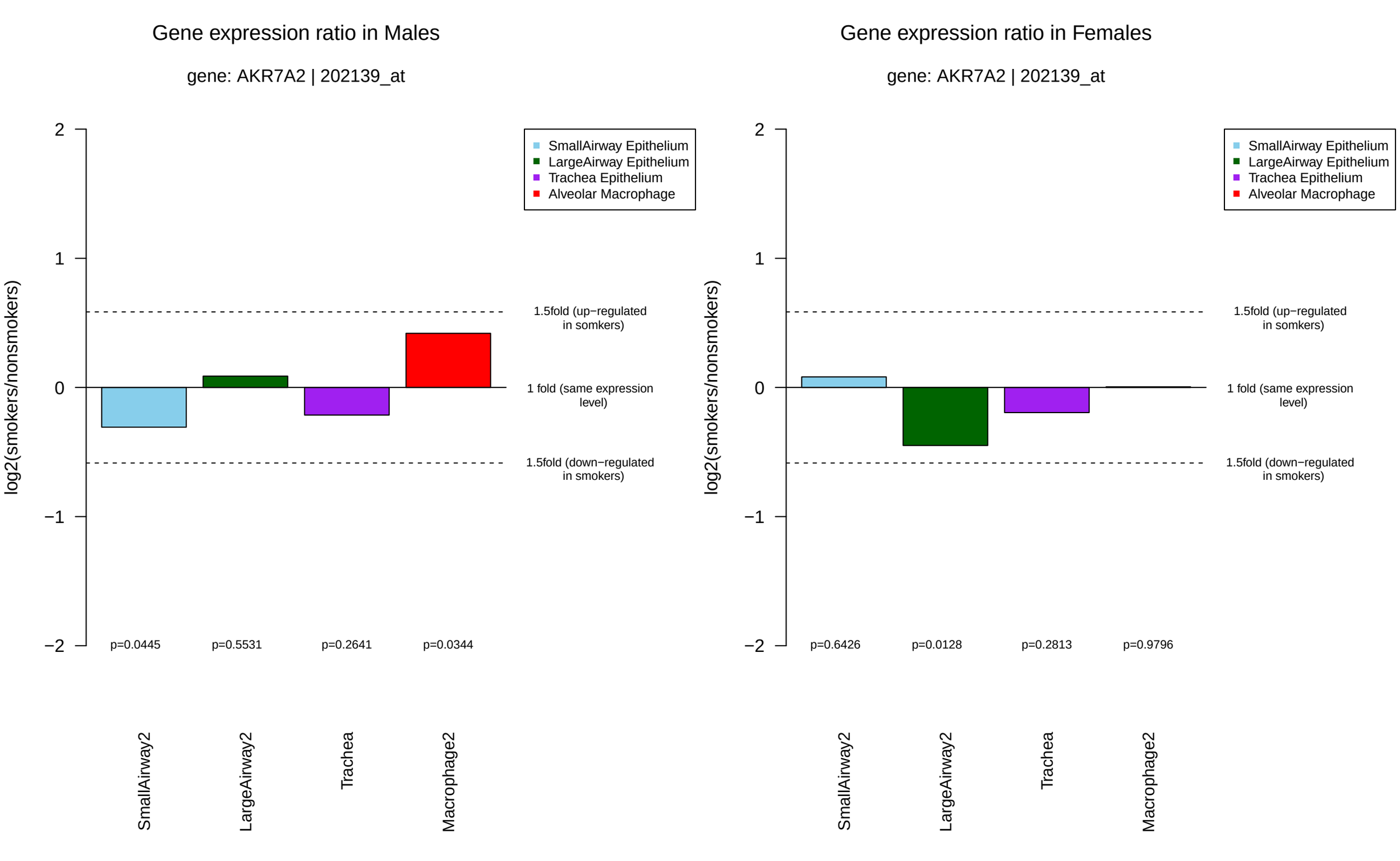
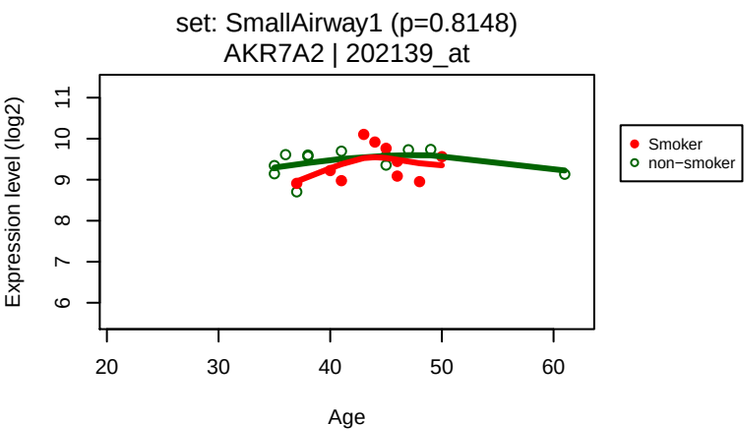
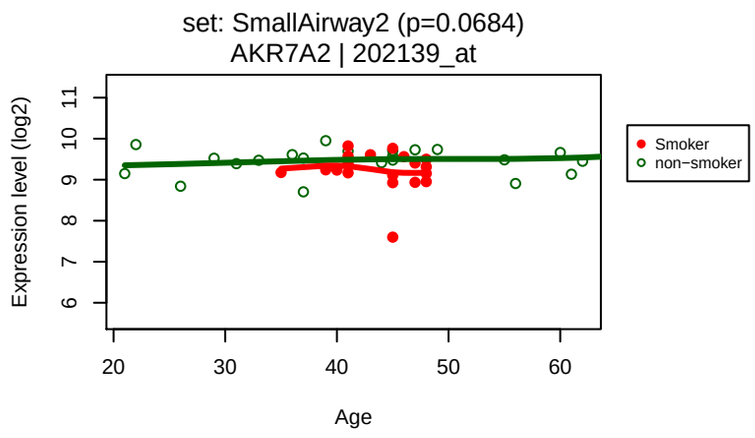
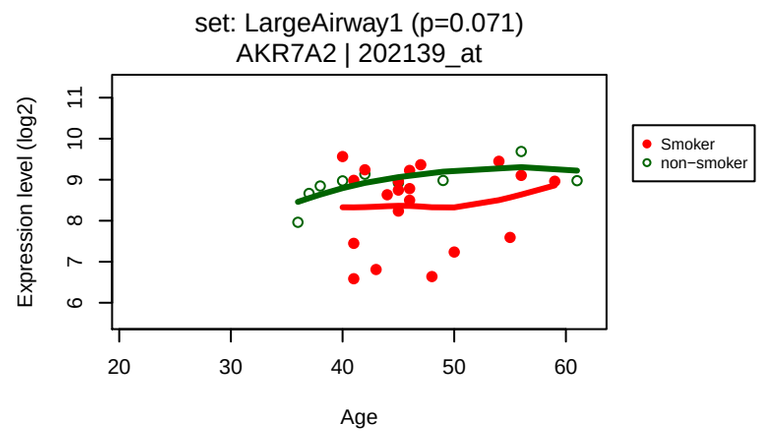

Gene expression of smokers and nonsmokers
gene: AKR7A2 | 202139_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8148
p=0.0684
p=0.2905
p=0.071
p=0.2809
p=0.1435
p=0.0955
p=0.0539
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

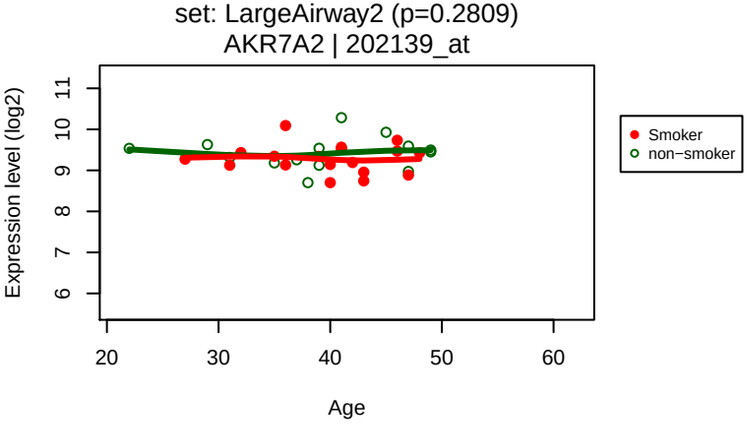

Gene expression ratio (smokers/nonsmokers)
gene: AKR7A2 | 202139_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8148
p=0.0684
p=0.2905
p=0.071
p=0.2809
p=0.1435
p=0.0955
p=0.0539
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

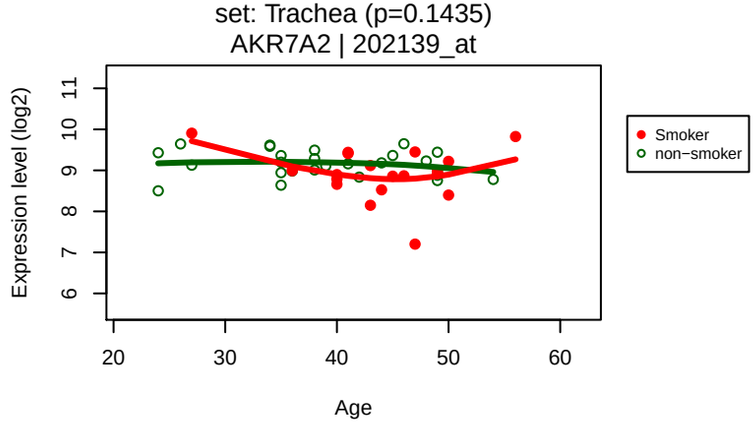

Gene expression ratio in Males
gene: AKR7A2 | 202139_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0445
p=0.5531
p=0.2641
p=0.0344
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKR7A2 | 202139_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6426
p=0.0128
p=0.2813
p=0.9796
SmallAirway2
LargeAirway2
Trachea
Macrophage2

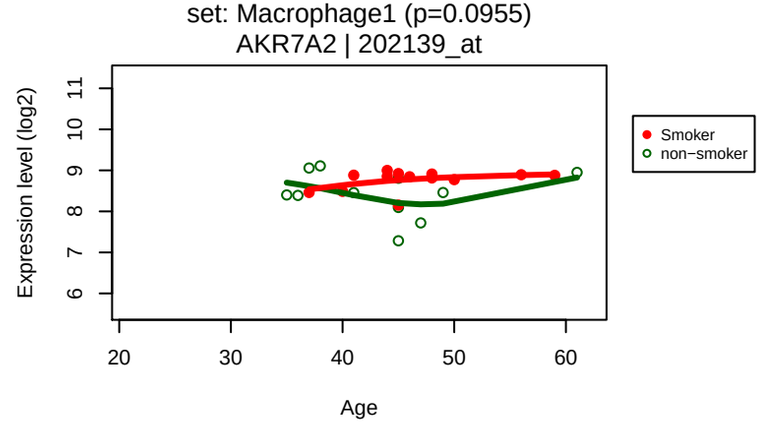

set: SmallAirway1 (p=0.8148)
AKR7A2 | 202139_at
Expression level (log2)
Age
Smoker
non−smoker

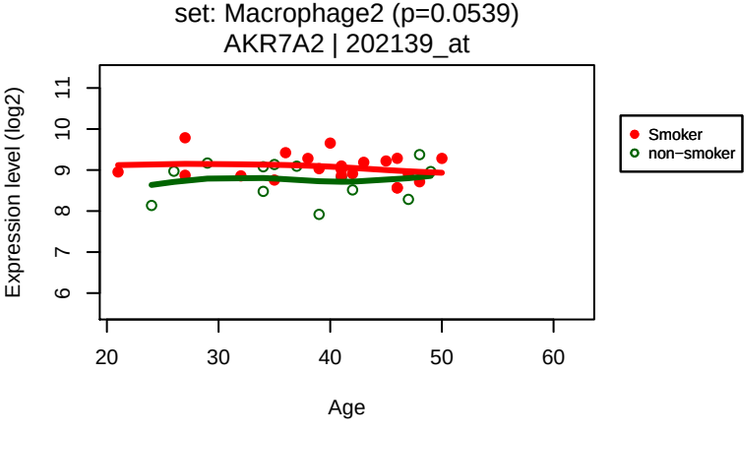

set: SmallAirway2 (p=0.0684)
AKR7A2 | 202139_at
Expression level (log2)
Age
Smoker
non−smoker

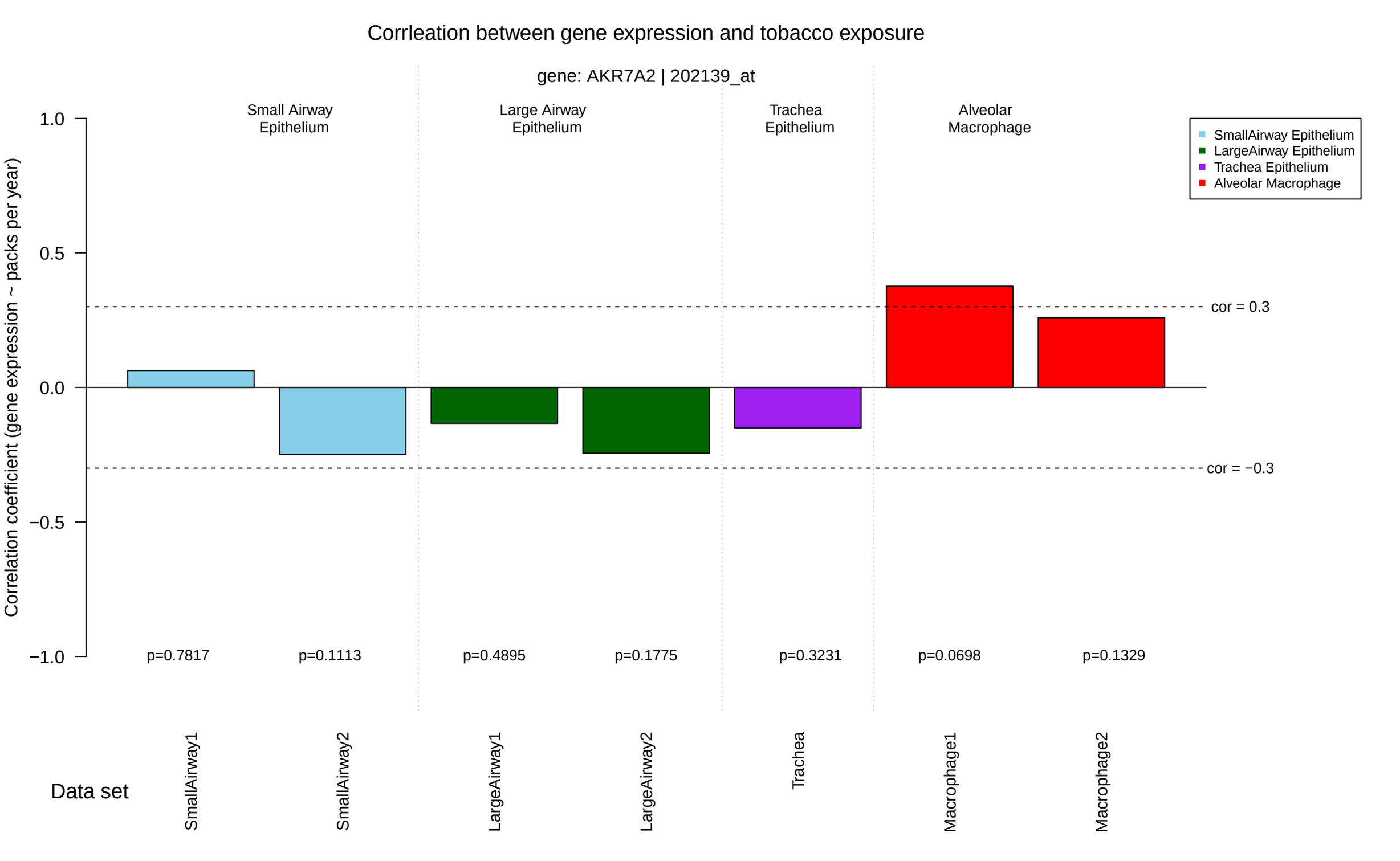

set: LargeAirway1 (p=0.071)
AKR7A2 | 202139_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2809)
AKR7A2 | 202139_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1435)
AKR7A2 | 202139_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0955)
AKR7A2 | 202139_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0539)
AKR7A2 | 202139_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKR7A2 | 202139_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7817
p=0.1113
p=0.4895
p=0.1775
p=0.3231
p=0.0698
p=0.1329
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2